POLICY BRIEF

Can Food Science and Technology Become a Leading Sector?

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In terms of long-term growth prospects, it cannot depend solely on foreign direct investment, the development of domestic firms can contribute to self-growth. The development of domestic firms faces many problems which are financial, technical, technological, low productivity, unable to fix costs. The development of the food industry to be a leading sector in the future as well as to take advantage of agriculture is something that decision-makers have to make up. This paper examines the positive impact of the agricultural sector on the development of food science and technology, the role of R&D in the promotion and development of the food sector, and possible policy responses.

Introduction

Over the last 20 years, Cambodia's economy has sustained an average growth rate of 8%, making it one of the fastest-growing economies in the world. Economic growth has been divining primarily by garment and footwear exports, commercial and residential construction, tourism, and agriculture.

But this successful growth has been amplified by recognition of openness to trade and flows, and stimulated by preferential trade treatment, and significant official development assistance (ODA), foreign direct investment (FDI) inflows 7.9% of GDP on average in 2005-2015, among the highest in the world (Work Bank, 2017). Because domestic firms do not have the capacity to produce and sell their goods and services in domestic and international markets. There are many reasons for this because domestic firms do not have enough budget to do business, low productivity, non-specialized (lack of skills and techniques), low technological innovation, unable to distinguish fixed costs and low production quality.

While underdeveloped domestic firms raise uncertainties about the future of economic growth. FDI still contributes to GDP growth but cannot depend on it. It is necessary to develop domestic firms to be able to produce quality goods and services that use advanced technologies and techniques with less time and more production. However, it is not possible to develop domestic firms in all sectors, priority sectors must be identified for development. So, what is the best sector to lead Cambodia's future economic growth?

Leading sectors are fundamental drivers of economic growth in a country that contributes significantly to GDP (Mardalena, Adji, Suhel, & Andaiyani, 2019; Reuveny & Thompson, 2001). Can food science and technology become a leading sector for Cambodia's future economic growth? Good policies can help develop domestic firms. In the development of food science and

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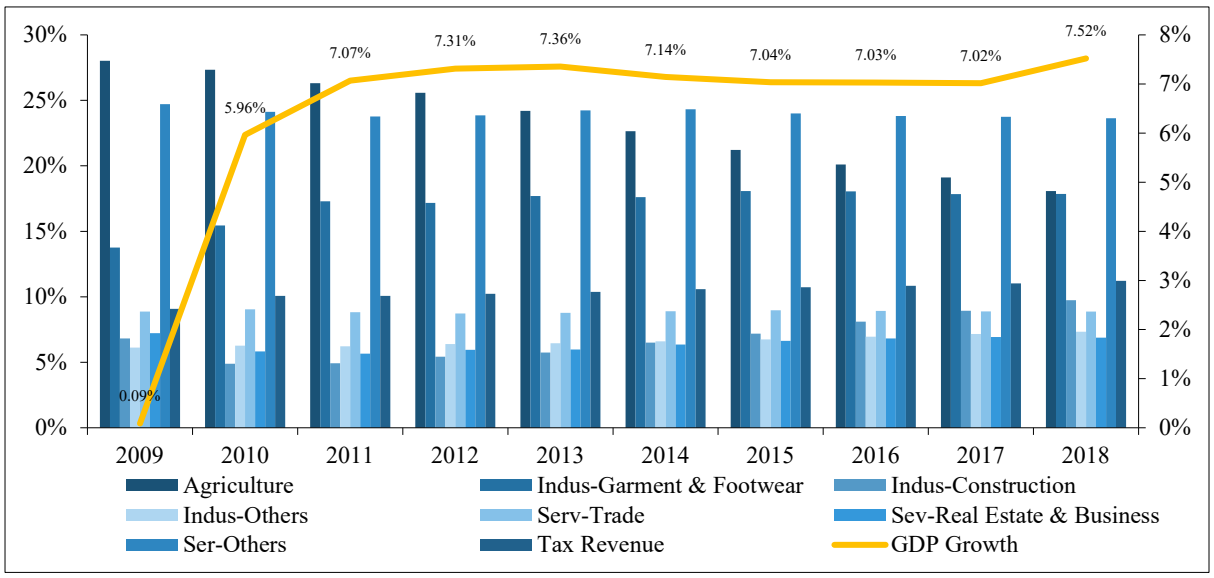
technology, more support for R&D from governments and donors, financial and taxes support, improving digital and physical infrastructure is the important thing to think about.

Leading Sectors Drive Economic Growth

Cambodia's economic growth is driven by three leading sectors, agriculture, industry and services. According to the Ministry of Economy and Finance, the share of GDP in the agricultural sector has declined over time, but it remains crucial for Cambodia's economy, representing about 28.02% of GDP in 2009, has declined to 22,65% in 2014 and the share of GDP will continue to decline to 18.08% in 2018. However, agricultural income to GDP has increased over time to 7,994,674 million riels in 2009, 9,401,243 million riels in 2017 and 9,527,348 million riels in 2018. The industry sector contributed about 26.60% to GDP in 2010, 30.71% in 2014 and 34.94% in 2018. Garment and footwear are a subsector in the industry sector contributed to GDP about 5,631,251 million riel equals 17.30% in 2011, 18.05% in 2016 and 17.85% in 2018 that 9,408,796 million riels. In addition, the service sector contributed about 39% to GDP in 2010, 39.61% in 2015 and 39.40% in 2018. Trade, transport and communications, real estate and business, as well as other services, are important subsectors in the services sector that contribute to GDP (Figure 1).

Figure 1: Percentage Contribution of Leading Sectors to Cambodia's GDP in 2009-2018



Source: Ministry of Economy and Finance 2018

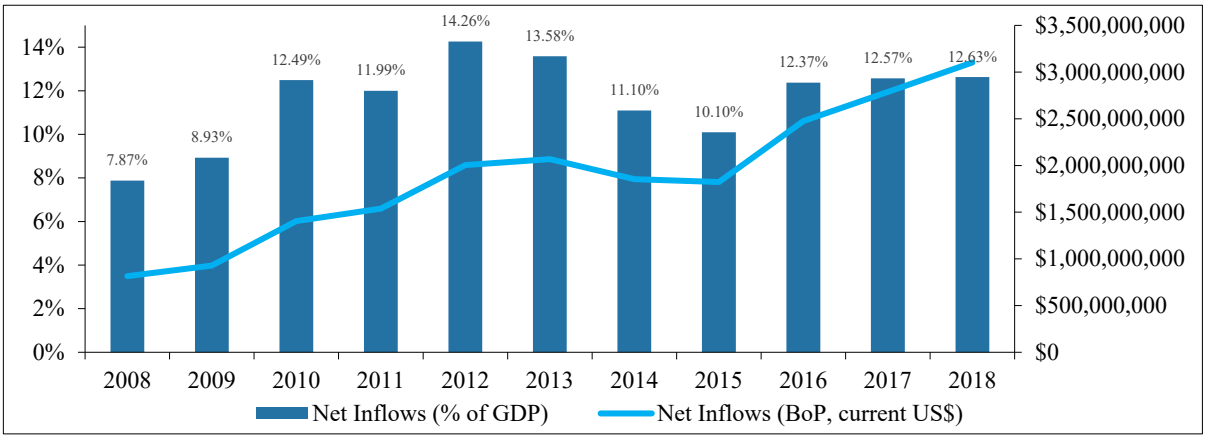
The percentage contribution of leading sectors varies from year to year, but at the same time, income by sector to GDP has increased year by year. However, Leading sectors will evolve over time, therefore, the policy development framework must target giving equivalent opportunity to all sectors, and this can best be accomplished in a structure of impediment evacuation than in that of particular favors (Chandra, 2006).

Recording in the World Economic Forum, Cambodia is ranked 107th out of 141 countries in terms of domestic competition performance, 121st out of 141 in terms of R&D performance, 106th in terms of infrastructure performance, 102nd in terms of innovation capacity, ranked 71st in ICT adoption and 120th in terms of skill performance at 141 countries (Schwab, 2019). The slow development of R&D, skills, ICT, infrastructure are the challenges for Cambodia in the development of domestic firms and the capture of foreign investment.

Foreign Direct Investment in Cambodia

Good experience in foreign direct investment in China, including multinational corporations (MNCs) are the main sources of advanced technological innovation and dissemination (He & Zhao, 2011). In fact, foreign direct investment is critical for developing and emerging market countries. In the real sector, private investment continued in the construction, real estate, and tourism sectors largely driven by FDI inflows, mounted to new highs. Maintained by the construction progress, key manufacturing activities such as building material, furniture, metal, and plastic products have also quickly succeeded (World Bank, 2019). FDI net inflows have grown by 7.87% or US$ 0.82 billion in 2018, it continues increased to 13.58% or US$ 2.07 billion in 2013 and continue increased about US$ 3.1 billion or 12,63% in 2018 (Figure 2).

Figure 2: Foreign Direct Investment, Net Inflow in 2008-2018



Source: Cambodia Economic Update (World Bank, 2019)

In recently, Chinese FDI inflows account for a large proportion of the recent surge in the inflows. As FDI inflows are coming largely from China, risks are associated with an unexpected slowdown in Chinese FDI (World Bank, 2019). A slowdown or reversal of FDI inflows, particularly Chinese FDI, will significantly affect private investment and growth. In addition, spillovers from a large economy, such as China, can occur both through trade and financial channels. The channels, namely FDI and tourism through which growth spillovers from large economies, such as China, can affect the Cambodia growth rate. However, finding clearly specific and determines the main sector is good opportunities for the development of domestic firms to promote production, export and growth. I see that the food sector is an important sector to develop.

What is Food Science and Technology?

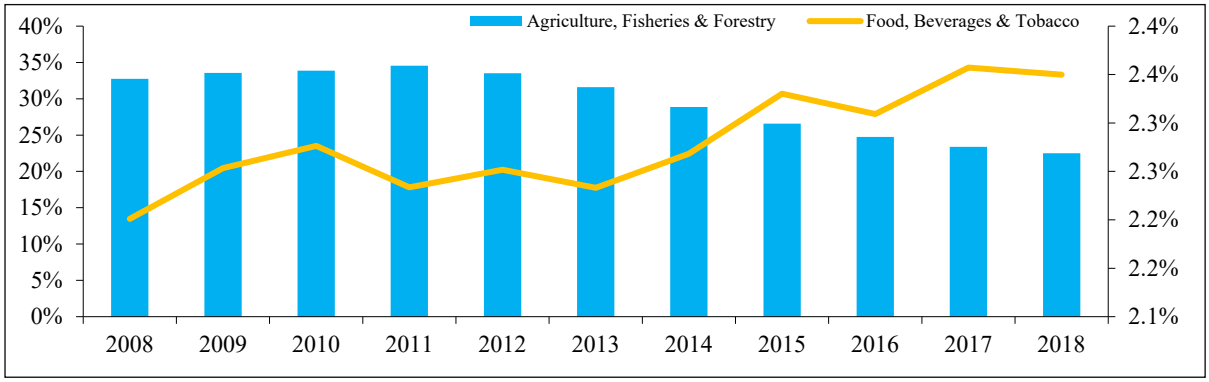
Food science is elementary sciences and manufacturing to study the basic chemical, physical properties, biochemical nature of foods and food processing. Food technology is the utilization of the information generated by food science in the choice, protection, processing, packaging, and distribution, as it influences the consumption of safe, nutritious and healthy food (Potter & Hotchkiss, 2012). Contemporary food science and technology have contributed significantly to the success of this modern food system by integrating biology, chemistry, physics, engineering, materials science, microbiology, nutrition, toxicology, biotechnology, genomics, computer science, and many other disciplines to solve difficult problems, such as resolving nutritional deficiencies and enhancing food safety (Floros et al., 2010).

Nowadays, they have been applied technology systems in the process and operation of the food industry including mechanical operation, heating, refrigeration and freezing, dehydration, fermentation, smoking, extrusion, packaging, microwave heating, ohmic heating, high-pressure processing, and pulsed electric fields (Burda, 2012; Floros et al., 2010; Sarkar & Costa, 2008). Many scholars believe that the future of the agriculture system and food science will be mainly by the trajectory of population and demographics, obtainability and type of energy resources, and effect of climate on available land, water, and air quality.

Can Food Science and Technology Become a Leading Sector?

Agriculture has been a leading economic growth in the past, current growth and may be in future growth. Crops, livestock and poultry, fisheries, forestry and logging are the main sub-sectors contributing to the growth of agriculture. At the same time, the food industry is less developed, contributing 2.2% to GDP, while agriculture contributed 32.8% in 2008. In 2018, it contributed about 2.3% of GDP, while agriculture contributed 22.5%. (Figure 3). It is the best that uses agricultural production to the final product by developing the food industry. The final product is more expensive than intermediate products and raw materials. An agricultural production price is twice as low as the global production of the food industry (Urban, 2014).

Figure 3: Percentage Contribution of Cambodia's GDP in 2008-2018



Source: Ministry of Economy and Finance 2018

As a result of the transformation in social and economic development, many farm workers have moved to other sectors. It is observed that the labour force in the agricultural sector has been remarkably reduced. It was 57.6% of total labour in 2009 and felled to 54.2% in 2010. In particular in 2013, it downed to 48.7%. In 2016, this trend is downed to about 36.4% and 30% in 2019. The trend of migration from rural to urban and overseas areas had increased. The labour force in the industrial sector is only at 16.9% in 2011 and increased by 19.9% in 2013, reaching 26% in 2019. The service sector workforce was 37% in 2016 and increased to 42% in 2019 (ILO, 2019). In this sense, a new workforce in industry with a work history in agriculture can have a positive effect on food development.

In the context of long-term economic growth and sustainable development, it depends on increased demand for manufactured products through agricultural development (Acevedo, Mold, & Perez Caldentey, 2009). Health needs have made people love organic foods that are food security, reasonable price, taste, freshness and good appearance (Shafie & Rennie, 2012). Therefore, it may be positive for Cambodia that the development food sector should become a leading sector in the future. Much critical thinking by researchers has assumed that the food

processing industry has highlighted the strategic role of food processing for developing countries in the context of globalization (Wilkinson, 2012). The demand side and the supply side of food products bring an ever-increasing level of competitiveness (Sarkar & Costa, 2008). However, applying science and technology within the food industry has allowed food production in enough quantities to meet the needs of humans in society, as it has evolved.

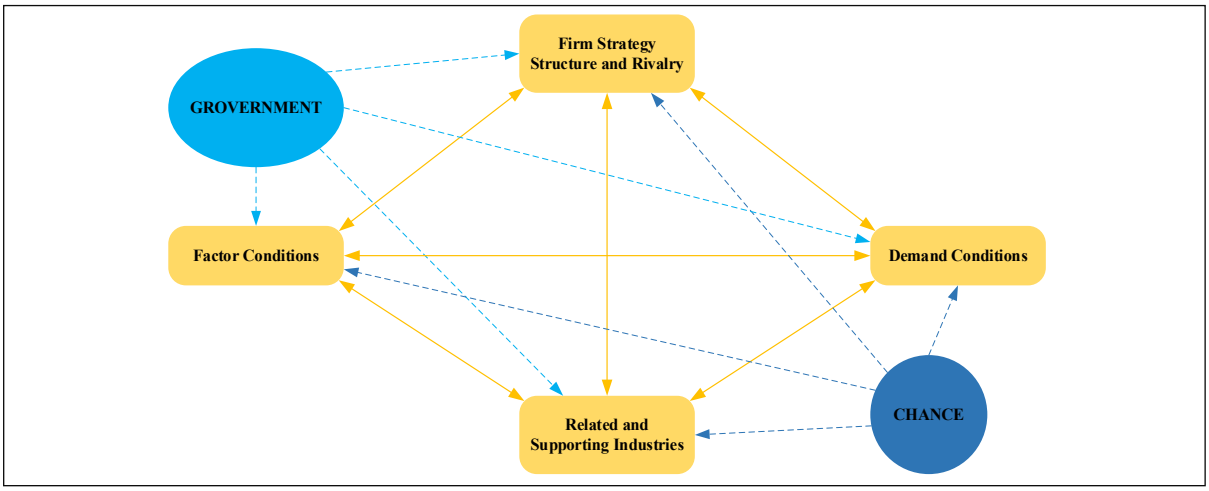
The effects of economies of scale, the new competitive environment, minimum quality, and non-traditional perspectives are the opportunities and challenges faced by SMEs in food processing and related activities. At the same time, integration of new activities and new actors into existing SMEs and new entrants like new SMEs with the advance in technical and technological to boost production in the food sector (Wilkinson, 2012).

Government's Role in Promoting the Leading Sectors

While the national development strategy must be equitable across all sectors, the government must also classify and identify the significance sector to develop for long-term growth. A leading sector is that it is a fundamental industrial or a new industry, commercial or financial technology that is increasing faster than other sectors within a major new technology. Leading sector contention is explicit about which new technological innovation generally liable for development.

A leading sector policy strategy is justified to development condition in a country has left from low growth equilibrium to a higher growth one, and to change employer from low income to areas where salaries are higher (Chandra, 2006; Currie, 1974). Many economists have proposed that governments play such an important role in supporting and assisting industry and employers through national policies to contribute straight to the competitive performance of target industries or firms (Porter, 1993), see (Figure 4).

Figure 4: Determinants of National competitive Advantage



Source: The Competitive Advantage of Nations, (Porter, 1993).

Recently, information technology (IT) was a foundation of US political economic leadership in the 20th century. In the 19th century, many researchers would agree that cotton, textiles, iron production, steam engines, railroads or steel, electricity, cars, semiconductors, aerospace were prominent foundations of British political-economic leadership (Crafts, 2004; Reuveny & Thompson, 2001). For another example, Singapore's national strategy focuses on R&D to develop technology and innovation, which spends about 2.1% of GDP in 2002. Malaysia's public

expenditure in 2004 represented about 0.7% of GDP for R&D in information and communications technologies (Monroe, 2006). Both Singapore and Malaysia structure innovation development share numerous likenesses, therefore depended on multinational firms to lead technical and technological development and innovative activity, but now the national policy in both countries is presently starting to underscore indigenous ability advancement to catch progressively financial and innovative overflows nationally. Activities in the two nations are being acquainted that look for improve inventive capacities by manufacturing neighborhood, local, and worldwide channels that encourage streams of information and information.

R&D Role in Promoting Long-Term Economic Growth

R&D is one of the most important operations of technological innovation to create new solutions in food production, the development of new foods and new conservation methods, packaging and labelling of food products (Burda, 2012). Innovation initiative leads to economic growth in a country, disseminates to the rest of the world, and stimulates the development of the world economy (Crafts, 2004; Currie, 1974; Lambert, McNamara, & Garrett, 2006; Modelski & Thompson, 1996; Monroe, 2006; Porter, 1993; Reuveny & Thompson, 2001; Sarkar & Costa, 2008). In example, advances in increased investment in technological innovation contribute to Japan's strong global competitiveness, which is characterized by low physical costs and flexible manufacturing structures (Matsuhashi & Togashi, 1988).

The statistical analyses found that firms with persistent R&D commitment have 13% higher labour productivity than non-R&D firms, and 9% higher productivity than firms which make occasionally R&D efforts, controlling for differences in past labour productivity. Moreover, a steady R&D strategy corresponds to around 2% higher growth rate in productivity (Börje & Hans, 2010). Advance in technological infrastructure, legal policies, and a good technological environment are very important to R&D activities. Both developing and developed countries cannot fully provide these factors – to maintain a strategic distance from its own natural weaknesses, building up R&D organizations in a decent situation might be a lot of proficient and successful than in the claim nation (He & Zhao, 2011).

Despite efforts, in the situation of small and medium-sized enterprises (SMEs), they are unable to distinguish fixed costs, non-specialization, low productivity and low production quality from large firms. In contrast, large manufacturing firms have progressed in a competitive advantage over small firms by reducing production costs, dropping time production that increases productivity, industry expertise, and a stable market structure because there are investments in R&D (Jung & Kwak, 2018; Kafouros, Wang, & Lodorfos, 2009 ; Porter, 1993; Reuveny & Thompson, 2001).

The limitation of the cash flow from SMEs is a consequence that they cannot improve their labour force, capital, and technical progress (Modelski & Thompson, 1996; Peretto, 1999). Thus, the government has an important role to play in helping domestic firms improve their R&D activities. However, a company created from a small capital and a small range of intermediate goods and after R&D was created with the aim of discovery innovation, reducing production costs and increasing commercial convenience, it is when the company makes an income by selling its products (Peretto, 1999).

Conclusions and Policy Recommendations

Development policies of national firms or SMEs help to promote production, technological innovation, new opportunities for workers, improved exports and enhanced long-term growth. It also reduces the impact of foreign investment on the country, because in the future it will be able to promote growth through domestic firms.

Thus, food science and technology become the leading sector for the future, it depends on the government's development policy. Cambodia has the potential of agriculture, so it is a very good thing to encourage the development of the food sector. I have seen significant policies in the improvement and development of the food sector in Cambodia become the leader in food science and technology.

a) **R&D in food science and technology,**

The Government of Cambodia should establish a National Research and Development Center. In the national strategy to improve the food industry has become a leading sector to promote growth in the future so that this central center must work closely in R&D to discover new technological innovations, new solutions, a new production method, a new management technique, policy review, legislation and programs to assess their practices and progress in the development of the food industry. In addition, it has a role in finding the food market and food international trade. Work closely with development partners and domestic firms, which in the project. At a minimum, the government must spend about 1% to 1.5% of GDP to the development of R&D. Furthermore, to promote and advance R&D in the food sciences, this center must be a partnership with other institutes, universities in the country, region, and world to collaborate in the exchange of experience and expertise.

b) **Food science and technology project**

All the technical, technological innovations and everything that has been found by the R&D center experts must be applied in the food industry. In the development of the food industry, it is necessary to identify specifically the number of food firms per stage of the policy framework. Experts are responsible for presenting and describing their new innovation for domestic firms and also have an important role in controlling, monitoring and evaluation of the firms' performance that applied new innovation found by the R&D center. In addition, they are consultants, trade facilitation and guidance firms to implement what they find and what firms expect.

c) **Promote and improve financial and fiscal support**

In order to achieve the long-term development objectives of the food sector, it is essential that the government has funds to support SMEs. In December 2017, the Cambodian Prime Minister recently launched an initiative to create a bank for SMEs in the hope of stimulating agro-firms and SMEs linked to foreign direct investment, tourism, and technology startup with an initial capital of US$100 million (Kimsay, 2019). This budget is so inadequate that the government should add more from year to year and strategically. It is hoped that the SME bank will be launched at the end of 2019, despite this, the SME bank should be implemented in a transparent manner, without corruption or other irregularities in lending to SMEs. To facilitate and promote the food science sector, the government should be in favor of taxation (specific tax law), facilitating taxation. At the same time, the R&D center's experts also helped to verify tax reviews, audits, financial statements, and fiscal facilitation.

d) The quality of digital and physical infrastructure

Extensive information and communications technologies (ICT) and reliable broadband service, with the ability to distribute to all types of computers, databases, heavy search tools, and access communications services, data services, software, and hardware. Improve the competitive environment in the telecommunications industry to ensure cheaper mobile and internet services. Prepare to have a submarine fiber optic cable and a satellite system by itself. The government must provide electricity everywhere and increase its own energy sources to help lower domestic electricity prices. In this regard, lower energy and logistics costs, the availability of a skilled workforce and improved supply chain links will be essential to remain competitive.

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